

Dhruva Kalyani



RESEARCH EXPERIENCE/EXPERIENCE

PLASTRO REU

May 2026 - Present

- 10-week research internship at West Virginia University
- Mentored by Dr. Emmanuel Fonseca
- Extracting the timing solutions for CHIME-observed pulsars
- Compiling analysis and results for future CHIME data release

Green Bank Observatory REU Student

May 2025 - August 2025

- 11-week research internship at the Green Bank Observatory
- Mentored by Dr. Jacob Turner
- Studied the scintillations of pulsar B1133+16 to better understand the ISM

Peter Livingston Scholars Program

October 2025 - January 2026

- Analyzing both the HI Absorption and Emission spectra of the SMC
- Utilizing Gaussian Decomposition and the process of Radiative Transfer

Green Bank Observatory Observer Training

November 2025

- 4-day training program at the Green Bank Observatory on using the Green Bank Telescope
- Mentored by Dr. Jacob Turner and Brenne Gregory in using the GBT for pulsar observations

NANOGrav Conference

November 2025

- Attended the NANOGrav Collaboration Meeting at the University of Montana

AAS Winter Meeting

January 2026

- Attended the AAS Winter Meeting in Phoenix, AZ

EDUCATION

2024 - Present Sophomore at **University of Wisconsin - Madison**

(GPA: 3.540/4.0)

- Majors: Astrophysics, Physics, Data Science
- Relevant Coursework: Intermediate courses in Astrophysics, Physics and Data Science

PUBLICATIONS

Kalyani, Dhruva and Jacob Turner (June 2026). “Determining the viability of a simple power law in strong scattering regimes with B1133+16”. In: Preparation/Drafting.

ORGANIZATIONS

Astronomy Club at UW-Madison Event Planner

August 2025 - Present

- Member of the Astronomy Club executive board
- Spearheading astronomy-related events for members to participate in

Milwaukee Astronomical Society

May 2022 - Present, New Berlin

- Elected Board Member and Appointed Keyholder in 2023
- Provided tours of the telescopes at open houses and volunteered

OUTREACH

Physicists Inspiring the Next Generation (PING) Mentor

July 2025 - July 2025

- Mentor for 18 rising high school freshmen over a 2-week camp at the Green Bank Observatory
- Led a group of 5 students in their research project for the camp

Wonders of Physics Scholar

October 2024 - May 2025

- Held physics-related outreach events for members of the community
- Committed to 30 - 40 hours of service

JOB EXPERIENCE

Physics Department Lab Assistant

January 2025 - Present

- Organizing the lab equipment for instructional physics labs
- Cleaning storage rooms and lab rooms

Kumon Tutor

July 2021 - July 2024

- 5 hours a week
- Entered scores of Students' work on a computer
- Supported Students in Homework and Classwork

SKILLS

Languages:	English, Spanish, Kannada
Programming Languages:	Python, JavaScript, R, HTML, UNIX/LINUX
Scientific Python Packages:	Numpy, Scipy, Astropy, Matplotlib, Scintools, PINT
Software Packages:	L ^A T _E X, Jupyter Lab/Notebook, Visual Studio Code

AWARDS

First Place Astronomical League Horkheimer Parker Imaging Award	July 2023
First Place Astronomical League Horkheimer D'auria Service Award	July 2024
Wells Fargo Employees' Dependent Children Scholarship Recipient	July 2024
Peter Livingston Scholars Program Award Winner	October 2025
Laufman Undergraduate Award	October 2025
Stencel Undergraduate Research Fellowship	June 2026